

Lecture 4 – Search Based Software Testing

AAA705: Software Testing and Quality Assurance

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2026 Fall

- Graph Coverage
 - NC / EC / k -PC / PPC – the k knob, prime paths for loops
 - Data-flow: ADC / AUC / ADUPC; subsumption is about guarantees
- Logic Coverage
 - PC / CC / CoC $\rightarrow$ determination $\rightarrow$ ACC (MC/DC), GACC / RACC / CACC
- Measuring Coverage – percentages, infeasible TRs, instrumentation
- Is Coverage a Good Criterion?
 - Necessary, not sufficient; low coverage is reliable bad news
 - Uses: a **filter**, a **search gradient**, not a quality measure
- Coverage Beyond Code – neuron coverage, spec coverage

The fuzzer's signal was **one bit** per branch: taken or not. It kept an input when a bit was **new**. Last time we called that coverage as a **search gradient**.

- One bit has no **direction**. For **if** ($x == 42$) every input but one looks the same – a **needle in a haystack**.
- A **distance** does: $|x - 42|$ says which way to move, and how far.
- Today: turn each **test requirement** into a distance and let a **search algorithm** drive it to 0. First the algorithms, then the testing.

1. Search Based Software Engineering (SBSE)
2. Fitness Landscape
3. Local Search
 - Simulated Annealing
 - Tabu Search
4. Genetic Algorithms
 - Representation and Operators
 - Selection
5. Bio-inspired Algorithms
 - Particle Swarm Optimization (PSO)
 - Ant Colony Optimization (ACO)
6. Search Based Software Testing (SBST)
 - Fitness for Branch Coverage
 - Alternating Variable Method (AVM)
 - From One Branch to a Test Suite

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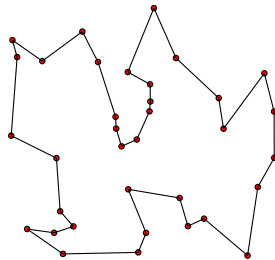
Ant Colony Optimization (ACO)

6. Search Based Software Testing (SBST)

Fitness for Branch Coverage

Alternating Variable Method (AVM)

From One Branch to a Test Suite



Travelling salesman: visit every city once and return. With 20 cities there are $19!/2 \approx 6 \times 10^{16}$ tours – enumeration is out.

- **Representation** – what a candidate looks like: a permutation of the cities.
- **Fitness function** – how good it is: the tour length.
- **Operators** – how to move to a neighbor: swap two cities.

Nothing about salesmen is left. Any problem written as these three parts – test inputs, test orderings, patches – can be handed to the **same** search algorithms. That is **search based software engineering**.¹

¹[IST'01] M. Harman, B. F. Jones. "Search-Based Software Engineering." [CSUR'12] M. Harman, S. A. Mansouri, Y. Zhang. "Search-Based Software Engineering: Trends, Techniques and Applications."

A **metaheuristic** is a problem-independent recipe for finding a **good enough** solution:

- **approximate** – no guarantee of optimality,
- **randomized** – two runs give two answers,
- **iterative** – try, measure, adjust, try again.

Every one of them balances two forces:

- **Exploitation** – a good solution probably has good neighbors. Look around it.
- **Exploration** – the part of the space nobody has visited may hold something much better. Go there.

Too much exploitation and the search sits in the first valley it finds; too much exploration and it is random testing again. The algorithms today are different ways to set this balance.

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From One Branch to a Test Suite

Find (x, y) with $x + y = 10$ for $0 \leq x, y \leq 10$ – but pretend we cannot solve equations. We may only **evaluate** candidates.

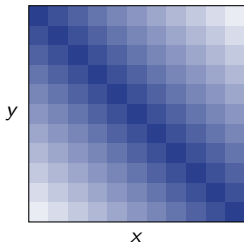
- **Representation:** a pair (x, y) of integers.
- **Fitness:** turn the equation into a distance to **minimize**,

$$f(x, y) = |10 - (x + y)|$$

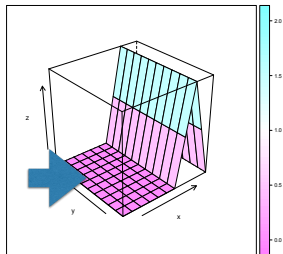
and $f = 0$ means solved.

- **Operators:** $x \pm 1, y \pm 1$.

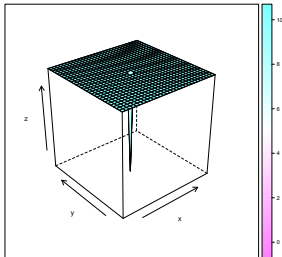
Plot f over every candidate and you get the **fitness landscape**: here a valley along the diagonal $x + y = 10$, sloping down from both corners.



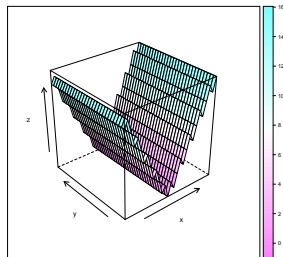
Three Shapes That Defeat Search



Plateau: all neighbors equal – no direction



Needle in a haystack: one good point, the rest flat

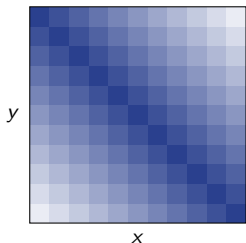


Rugged: many local optima – the nearest valley is rarely the deepest

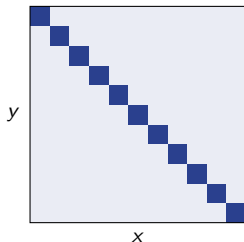
- On a plateau or a needle **no** algorithm beats random sampling: the fitness carries no information until you are already there.
- On a rugged landscape the algorithm matters – it decides how the search gets **off** a local optimum.

The Landscape Is Ours to Design

The landscape is not a property of the problem. It is a property of the **fitness function** we chose.



$f(x, y) = |10 - (x + y)|$ – a slope from
anywhere



$f(x, y) = [x + y \neq 10]$ – the fuzzer's
bit

- Same problem, same representation, same operators. Whether the search takes ten steps or forever was decided **before it started**.

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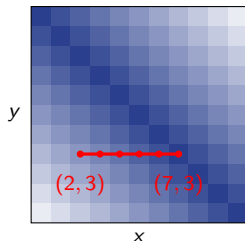
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Fitness for Branch Coverage

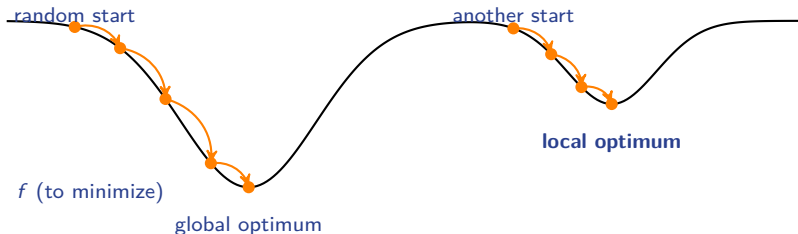
Alternating Variable Method (AVM)

From One Branch to a Test Suite

```
1:  $s \leftarrow$  a random solution
2: loop
3:    $s' \leftarrow$  the best of  $neighbors(s)$ 
4:   if  $f(s') \geq f(s)$  then return  $s$ 
5:    $s \leftarrow s'$ 
```



- From $(2, 3)$: $f = 5, 4, 3, 2, 1, 0$. Five moves, **20** evaluations. Random sampling needs about 11 tries to hit one of the 11 solutions out of 121.
- We **minimize** f throughout, so “ascent” means a **smaller** f and the search stops when no neighbor has one. Which improving neighbor to take: **steepest** (evaluate all, take the best), **first** improving (cheaper per step), or a **random** improving one.
- The operator sets the **radius**, not the size, of the search: 20 cities give 6×10^{16} tours, but any tour reaches any other in at most 190 adjacent swaps.



- A **local optimum**: no neighbor is better, and it is not the answer. Plain hill climbing stops here.
- Three ways out, and the rest of this section is about them:
 - **restart** from a new random point while budget remains,
 - **accept a worse move** sometimes – simulated annealing,
 - **remember** where you have been – tabu search.

```

1:  $s \leftarrow$  a random solution;  $T \leftarrow T_0$ 
2: while  $T > T_{\min}$  do
3:    $s' \leftarrow$  a random neighbor of  $s$ 
4:    $\Delta \leftarrow f(s') - f(s)$ 
5:   if  $\Delta < 0$  or  $\text{rand}() < e^{-\Delta/T}$  then
6:      $s \leftarrow s'$ 
7:    $T \leftarrow \text{cool}(T)$ 
    
```

	$T = 10$	$T = 1$	$T = 0.1$
$\Delta = 1$	0.90	0.37	0.00005
$\Delta = 5$	0.61	0.007	≈ 0

probability of accepting a **worse** move

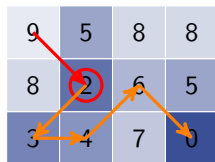
- Hot: almost a random walk (**explore**). Cold: hill climbing (**exploit**).
The **cooling schedule** moves from one to the other – linear $T_0 - \alpha t$ or geometric $T_0 \alpha^t$.²
- Logarithmic cooling $c / \log(t + d)$ has a theorem: with c large enough, the **probability** of sitting at a global optimum tends to 1 as $t \rightarrow \infty$.
No guarantee for any finite run – and it needs more steps than trying every solution, so nobody uses it.³
- The directed fuzzer we saw did exactly this: the weight on the distance to the target **grew over time**.

²[Science'83] S. Kirkpatrick, C. D. Gelatt, M. P. Vecchi. "Optimization by Simulated Annealing."

³[MOR'88] B. Hajek. "Cooling Schedules for Optimal Annealing."

Move to the **best neighbor that is not tabu** – even when it is worse.

The **tabu list** holds the last k positions visited; here $k = 3$.⁴



8-neighborhood

at	tabu	allowed neighbors	move
2	9, 2	3, 4, 5, 6, 7, 8, 8	→ 3, worse – best of 7
3	9, 2, 3	4, 8	→ 4
4	2, 3, 4	6, 7, 8 (2 is tabu)	→ 6, worse
6	3, 4, 6	0, 2, 5, 5, 7, 8, 8	→ 0, optimum

Hill climbing (red) stops at 2: all eight neighbors are worse.

- The search still looks only at neighbors and moves one step. What memory changes is **which neighbor counts as best**: at 4 the best neighbor is the 2 two steps back, and it is forbidden.
- k is the knob. With $k = 2$ the list at 4 is $\{3, 4\}$, so 2 is allowed and the search **cycles** 2, 3, 4, 2, 3, 4, ... – a loop of three needs three steps of memory. **Aspiration**: a tabu move is taken if it beats the best seen so far.

⁴[C&OR'86] F. Glover. "Future Paths for Integer Programming and Links to Artificial Intelligence."

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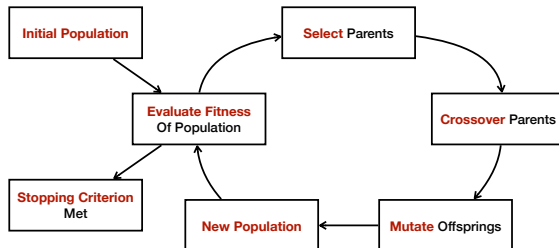
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Fitness for Branch Coverage

Alternating Variable Method (AVM)

From One Branch to a Test Suite



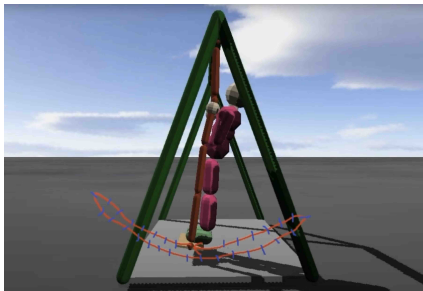
- Not one point but a **population**. Many regions of the space are sampled at once, and good **parts** of different solutions can be combined.⁵
- **Selection pressure** is the exploitation knob: the fitter an individual, the more children it gets. Too much and the population collapses onto one local optimum (**premature convergence**); too little and it drifts.
- **Elitism**: copy the best few unchanged, the best fitness never gets worse.

⁵[Holland'75] J. H. Holland. "Adaptation in Natural and Artificial Systems."

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>	—	2	5	4	6
<i>B</i>		—	3	6	5
<i>C</i>			—	2	4
<i>D</i>				—	3
<i>E</i>					—

	tour	length
P_1	<i>A B D E C</i>	20
P_2	<i>D B C E A</i>	23
P_3	<i>A C D B E</i>	24
P_4	<i>B D A C E</i>	24

- **Select** by tournament of two: P_1 beats P_3 , P_2 beats P_4 .
- **Crossover** for permutations: keep a **slice** of one parent, fill the rest in the **other's order**. Slice *B C E* from P_2 , then *A, D* as they appear in P_1 : *A B C E D*, length **16** – the optimum. The other child, *C B D E A*, has length 23.
- **Mutate**: swap two cities. *C B D E A* with $D \leftrightarrow A$ becomes *C B A E D* – also 16.
- Picking good parents and recombining their **pieces** is the **exploitation**; mutation is the **exploration** – the only source of material the population does not already have.



Video: GA riding a swing

- **Representation:** one swing cycle cut into 32 time steps, one bit each – 1 stand, 0 sit.
- **Fitness:** how high the swing gets.
- **Operators:** flip a bit, cut two bit strings at a point and swap the tails.
- Nobody told it **when** to stand. The population found the rhythm.

solution	encoding	crossover	mutation
a tour	permutation	keep a slice, fill the rest in the other parent's order	swap two cities
a program, a formula	syntax tree	swap subtrees	replace a node
a unit test	statement sequence	exchange statement sequences	add, delete, change a statement

- Crossover must produce a **valid** solution: one-point crossover of two tours duplicates cities, so permutations get their own operator.
- Mutation is the **only** source of new material; crossover only recombines what the population already has.
- The last two rows return later in the course: trees when formulas and programs are evolved, statement sequences when we generate unit tests.

Who gets to be a parent? Three individuals, fitness f to **minimize**:

	f	rank	$P_{\text{proportional}}$	P_{rank}	$P_{\text{tournament}} (k = 2)$
A	1	1	0.59	0.50	0.56
B	2	2	0.29	0.33	0.33
C	5	3	0.12	0.17	0.11

- **Fitness proportional** (roulette wheel): pick i with probability $\propto 1/f(i)$ – the wheel needs a score to **maximize**, so invert. Simple – but one very fit individual soon takes over the whole population, and $f = 0$ breaks it.
- **Ranking**: sort first, then pick by **rank** instead of fitness, $\propto N + 1 - \text{rank}$. A distance of 1 next to one of 100 is just first next to second, so the gap is capped.
- **Tournament**: draw k at random, keep the best. Only comparisons, so no inversion; a bigger k means more pressure. Most tools use this.

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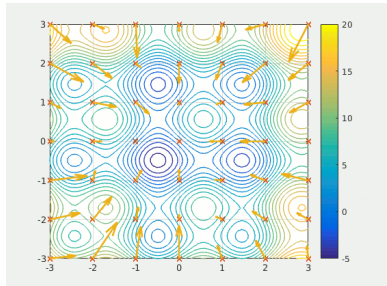
Alternating Variable Method (AVM)

From One Branch to a Test Suite

A flock of birds. Each bird remembers **its** best spot p_i and hears the **flock's** best spot g .⁶

$$v_i \leftarrow \textcircled{1} w v_i + \textcircled{2} c_1 r_1 (p_i - x_i) + \textcircled{3} c_2 r_2 (g - x_i) \quad x_i \leftarrow x_i + v_i$$

- $\textcircled{1}$ keep going $\textcircled{2}$ toward my best
 $\textcircled{3}$ toward the flock's best
- GA **competes**: the weak are dropped.
 PSO **cooperates**: nobody is dropped.



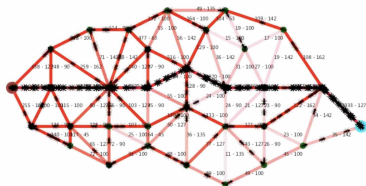
Animation

⁶[ICNN'95] J. Kennedy, R. Eberhart. "Particle Swarm Optimization."

Ants leave **pheromone** on the path they walk and prefer edges with more of it. Short paths get walked more, so they get more.⁷

$$p_{ij} \propto \tau_{ij}^{\alpha} \cdot (1/d_{ij})^{\beta} \quad \Delta\tau_{ij} = Q/L \quad \tau_{ij} \leftarrow (1 - \rho) \tau_{ij} + \sum \Delta\tau_{ij}$$

- Pheromone τ : **shared memory** of good edges.
- Evaporation ρ : **forgetting**, so the colony keeps exploring.
- Made for **paths on graphs**.



Video

⁷[SMC'96] M. Dorigo, V. Maniezzo, A. Coloni. "Ant System: Optimization by a Colony of Cooperating Agents."

	explores by	exploits by
hill climbing	random restarts	always taking a better neighbor
simulated annealing	accepting worse moves while hot	cooling
tabu search	forbidding recent moves	best allowed neighbor
genetic algorithm	population, mutation	selection, crossover
PSO / ACO	inertia / evaporation	global best / pheromone

- **No free lunch:** averaged over **all** possible fitness functions, every search algorithm performs exactly the same. An algorithm wins only where its moves match the **shape** of the landscape.⁸
- So the question is never “which algorithm is best” but “what does **our** landscape look like”. For testing, that is the next section.

⁸[TEC'97] D. H. Wolpert, W. G. Macready. “No Free Lunch Theorems for Optimization.”

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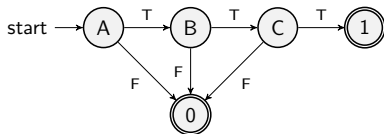
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From One Branch to a Test Suite

```
int f(int a, int b) {  
    if (a >= 10)           // A  
        if (b == 2 * a)    // B  
            if (a + b > 100) // C  
                return 1;  // target  
    return 0;  
}
```



- **Target:** the true branch of C. Which of the 2^{64} inputs (a, b) reach it?⁹
- **Representation:** (a, b) . **Operators:** ± 1 on one variable.
Fitness: how far the run stopped from the target.

⁹[TSE'90] B. Korel. "Automated Software Test Data Generation."

How close did a **false** condition come to being true?¹⁰

condition	distance when false	condition	distance when false
$a = b$	$ a - b $	$p \wedge q$	$d(p) + d(q)$
$a \neq b$	K	$p \vee q$	$\min(d(p), d(q))$
$a < b, a \leq b$	$a - b + K$	$\neg p$	push $\neg$ inward
$a > b, a \geq b$	$b - a + K$	if (flag)	K

- 0 when true. $K > 0$ (we use 1) keeps it positive when the operands are equal but the condition is false.
- Normalize into $[0, 1)$: $d/(d + 1)$.¹¹

¹⁰[ASE'98] N. Tracey, J. Clark, K. Mander, J. McDermid. "An Automated Framework for Structural Test-Data Generation."

¹¹[STVR'13] A. Arcuri. "It Really Does Matter How You Normalize the Branch Distance in Search-Based Software Testing."

$$fitness(t) = \underbrace{\text{approach level}}_{\text{ifs still ahead, beyond the one it left}} + \underbrace{\text{normalize(branch distance)}}_{\text{at the if where the path left}}$$

```
int f(int a, int b) {
    if (a >= 10) // A
        if (b == 2 * a) // B
            if (a + b > 100) // C
                return 1;
    return 0;
}
```

(a, b)	left at	distance	fitness
(3, 0)	A	$10 - 3 + 1 = 8$	$2 + 8/9 = 2.889$
(12, 0)	B	$ 0 - 24 = 24$	$1 + 24/25 = 1.960$
(12, 24)	C	$100 - 36 + 1 = 65$	$0 + 65/66 = 0.985$
(40, 80)	—	0	0

- Approach level = 2, 1, 0 for leaving at A, B, C. One `if` deeper always beats any progress on the current one.¹²

¹²[IST'01] J. Wegener, A. Baresel, H. Sthamer. "Evolutionary Test Environment for Automatic Structural Testing."
[STVR'04] P. McMinn. "Search-Based Software Test Data Generation: A Survey."

Hill climbing on one variable at a time.

```
1:  $t \leftarrow$  a random input
2: repeat
3:   for each variable  $x_i$  of  $t$  do
4:     exploratory: try  $x_i - 1$  and  $x_i + 1$ 
5:     while one of them improved  $f$  do
6:       pattern: keep going that way with
7:         steps 2, 4, 8, ... while improving
8:       then try  $x_i \pm 1$  again
9: until no variable improved      ▷ local optimum
10: restart from a new random input
```

- Exploratory moves find the **direction**.
- Pattern moves cross large ranges **fast**.

move	(a, b)	left at	fitness
start	(3, 0)	A	2.889
explore a	(2, 0) (4, 0)	A	▲ 2.900 ▼ 2.875
pattern $a + 2, +4$	(6, 0) (10, 0)	A B	▼ 2.833 ▼ 1.952
pattern $a + 8$	(18, 0)	B	▲ 1.973 back to (10, 0)
explore a	(9, 0) (11, 0)	A B	▲ 2.667 ▲ 1.957
explore, pattern b	(10, 1) ... (10, 15)	B	▼ 1.950 ... ▼ 1.833
pattern $b + 16$	(10, 31)	B	▲ 1.917 back to (10, 15)
explore, pattern b	(10, 16) (10, 18) (10, 19)	B	▼ 1.800 ▼ 1.667 ▼ 1.500
explore b	(10, 20)	C	▼ 0.986
explore a, b	(9, 20) (11, 20) (10, 19) (10, 21)	A B B B	▲ ▲ ▲ ▲ all worse

- **Stuck** at (10, 20) after 30 evaluations: every one-variable move breaks B. The solutions lie on the line $b = 2a$, and no one-variable move walks along it.

move	(a, b)	left at	fitness
start	(12, 90)	B	1.985
explore a	(11, 90) (13, 90)	B	▲ 1.986 ▼ 1.985
pattern $a + 2, +4, +8, +16$	(15, 90) ... (43, 90)	B	▼ 1.984 ... ▼ 1.800
pattern $a + 32$	(75, 90)	B	▲ 1.984 back to (43, 90)
explore, pattern a	(44, 90) (46, 90)	B	▼ 1.667 ▲ 1.667
explore a	(45, 90)	–	▼ 0

- **13** evaluations. But only starts with $b > 66$ succeed – about **one in three**.
- The fitness sees only the branch where the path **left**: while B is false it says nothing about $a + b > 100$.
- A solver that reads the **whole** path condition, $a \geq 10 \wedge b = 2a \wedge a + b > 100$, returns (34, 68) at once. That is the next lecture.

```
bool ok = check(x);    // computed from x, somewhere else
...
if (ok) target();      // branch distance: K when false, always
```

- A **boolean** has no distance. The landscape of `if (ok)` is a **plateau**, however much slope there was inside `check`.
- **Testability transformation**: rewrite the program **for the search only**, so the distance inside `check` flows out to the `if`.¹³
- The algorithm cannot fix a flat landscape. The fitness function can.

¹³[TSE'04] M. Harman, L. Hu, R. Hierons, J. Wegener, H. Sthamer, A. Baresel, M. Roper. "Testability Transformation."

So far, **one input** toward **one branch**. Two more ways to run it:

	what the search moves	what it minimizes
one branch at a time (AVM, so far)	one input	the distance to that one branch
whole test suite (EvoSuite)	a set of tests	the sum of the distances to all branches
many-objective (MOSA)	a set of tests	every branch its own distance, no sum

- One at a time has a trap: an **infeasible** branch never reaches 0 and eats the whole budget. Whole suite spends it where progress is possible – up to **188×** the coverage on 1,741 classes.¹⁴
- A sum hides **which** branch improved. MOSA keeps the best test so far for **each** branch – EvoSuite's default today.¹⁵

¹⁴[TSE'13] G. Fraser, A. Arcuri. "Whole Test Suite Generation."

¹⁵[TSE'18] A. Panichella, F. M. Kifetew, P. Tonella. "Automated Test Case Generation as a Many-Objective Optimisation Problem ..."

Random testing throws in inputs blindly and keeps what covers something new. Is all this machinery better?

- On **1,000 real Java classes** the GA covered **barely more** than random. Most branches in real code are like **if** (flag): the distance is **flat**, so there is nothing to climb.¹⁶
- Some branches fall to nudging one input step by step (AVM); others need a population that **combines** pieces (GA). Running **both** beat either alone.¹⁷
- **Bottom line:** the search helps only where the fitness has a **slope** to follow. On flat branches it is random testing that also pays to compute distances.

¹⁶[STVR'18] S. Shamshiri, J. M. Rojas, L. Gazzola, G. Fraser, P. McMinn. "Random or Evolutionary Search for Object-Oriented Test Suite Generation?"

¹⁷[TSE'10] M. Harman, P. McMinn. "A Theoretical and Empirical Study of Search-Based Testing: Local, Global, and Hybrid Search."

1. Search Based Software Engineering (SBSE)
2. Fitness Landscape
3. Local Search
 - Simulated Annealing
 - Tabu Search
4. Genetic Algorithms
 - Representation and Operators
 - Selection
5. Bio-inspired Algorithms
 - Particle Swarm Optimization (PSO)
 - Ant Colony Optimization (ACO)
6. Search Based Software Testing (SBST)
 - Fitness for Branch Coverage
 - Alternating Variable Method (AVM)
 - From One Branch to a Test Suite

- Dynamic Symbolic Execution

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